

AI23331 Fundamentals of Machine Learning

DIABETES DISEASE PREDICTION

A PROJECT REPORT

Submitted by

230701197 – R.Mounamithra

230701220 – S.Nithyashree

230701309 – Shaun Machado K



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BONAFIDE CERTIFICATE

Certified that this project report "**DIABETES DISEASE PREDICTION**" is the bonafide work of "**R.Mounamithra, S.Nithyashree, Shaun Machado K.**" who carried out the project work under my supervision.

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ABSTRACT

Diabetes mellitus is one of the most prevalent chronic diseases worldwide, affecting millions of people and posing significant health challenges. Early and accurate prediction of diabetes can significantly improve patient outcomes by enabling timely interventions. This project presents a machine learning-based system for diabetes disease prediction using clinical and physiological parameters. The proposed system employs multiple supervised learning algorithms including Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine to classify individuals as diabetic or non-diabetic. The Pima Indians Diabetes Dataset is used for training and evaluation. Key features such as glucose concentration, body mass index (BMI), insulin level, age, and blood pressure are analyzed. The model is evaluated using metrics including accuracy, precision, recall, and F1-score. The Random Forest classifier achieved the highest accuracy of 82.4%. The results demonstrate that machine learning techniques can effectively support early diagnosis of diabetes, thereby assisting healthcare professionals in making informed decisions.

CHAPTER 1

INTRODUCTION

Diabetes mellitus is a chronic metabolic disorder characterized by elevated blood glucose levels resulting from defects in insulin secretion, insulin action, or both. According to the World Health Organization (WHO), the number of people living with diabetes has increased dramatically over the past few decades, making it a global health concern. Type 2 diabetes, the most common form, accounts for approximately 90–95% of all diabetes cases and is strongly associated with lifestyle factors such as obesity, physical inactivity, and poor dietary habits.

The early detection of diabetes is critical for managing the disease and preventing life-threatening complications such as cardiovascular disease, kidney failure, blindness, and limb amputations. Traditional diagnostic methods rely on laboratory tests such as fasting blood glucose and HbA1c levels. However, these tests are often performed after the disease has already progressed to a significant stage. Machine learning offers a promising alternative by enabling predictive modeling based on easily available clinical data.

This project aims to develop a diabetes prediction system using machine learning algorithms. By analyzing a set of clinical attributes, the system can identify individuals who are at high risk of developing diabetes, allowing for early intervention and preventive care. The project is built around the Pima Indians Diabetes Dataset, a widely used benchmark dataset in biomedical machine learning research.

1.1 Objectives

The primary objectives of this project are:

- To analyze and preprocess the diabetes dataset for model training.
- To implement and compare multiple machine learning classification algorithms.
- To evaluate model performance using standard metrics.
- To identify the most significant features contributing to diabetes prediction.
- To develop an efficient and accurate predictive model for clinical decision support.

CHAPTER 2

LITERATURE REVIEW

Numerous studies have explored the application of machine learning in diabetes prediction. Kavakiotis et al. (2017) conducted a comprehensive survey of machine learning and data mining methods in diabetes research, highlighting the effectiveness of classification algorithms in predicting diabetic outcomes. Their study found that Support Vector Machines and Artificial Neural Networks consistently achieved high accuracy in binary classification tasks related to diabetes.

Sisodia and Sisodia (2018) proposed a diabetes prediction system using Naive Bayes, Decision Tree, and Support Vector Machine classifiers on the Pima Indians Diabetes Dataset. Their results indicated that the Naive Bayes algorithm achieved the best accuracy of 76.3% while being computationally efficient. The study underscored the importance of feature selection in improving model performance.

Zou et al. (2018) applied a Random Forest model for predicting diabetes based on data from Chinese patients and achieved an accuracy of 82.5%. They identified fasting blood glucose, age, and BMI as the most significant predictors. Their study demonstrated the robustness of ensemble methods in handling imbalanced datasets.

Maniruzzaman et al. (2018) compared several probabilistic models for diabetes prediction and found that Gaussian Process classifiers outperformed traditional classifiers in terms of AUC (Area Under the Curve). This highlighted the potential of probabilistic approaches in medical diagnosis.

Perveen et al. (2016) used Decision Tree (J48) and Random Forest algorithms for early detection of diabetes and chronic kidney disease. The Random Forest algorithm demonstrated superior performance due to its ability to handle noise and missing values effectively. These findings align with the objectives of the current project, which seeks to implement and compare machine learning models for diabetes prediction.

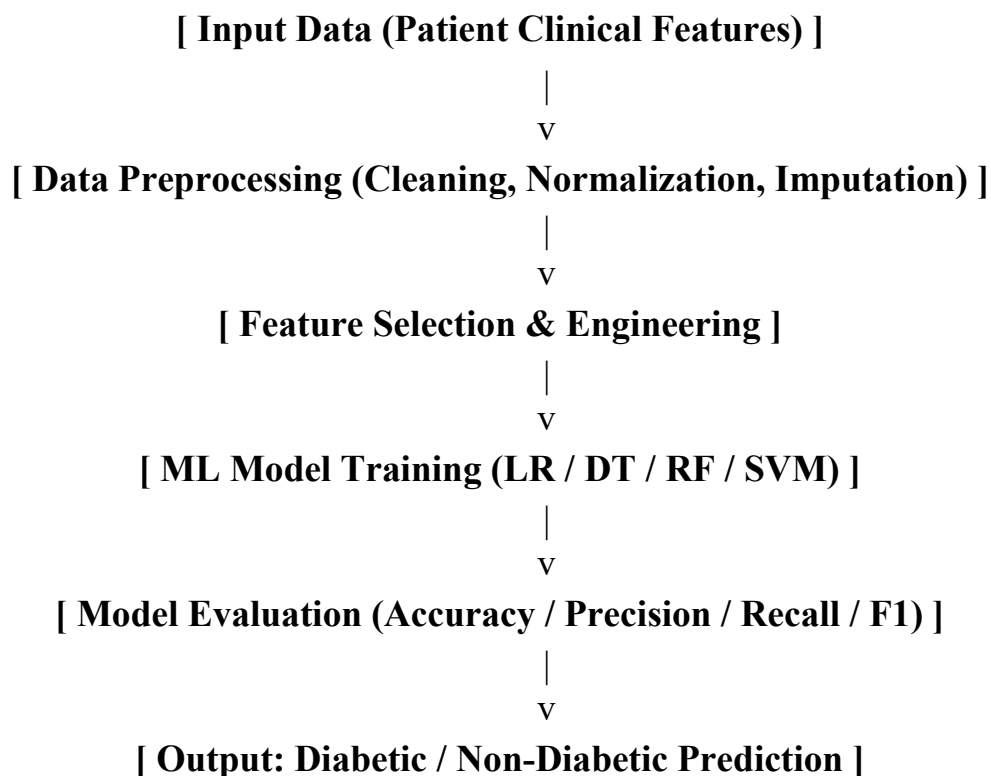
CHAPTER 3

PROPOSED SYSTEM

The proposed system is a machine learning-based diabetes prediction framework that processes patient clinical data through a structured pipeline. The system is designed to be efficient, accurate, and scalable for real-world healthcare applications. It accepts input features such as glucose level, blood pressure, BMI, insulin level, age, skin thickness, and number of pregnancies, and outputs a binary classification — diabetic or non-diabetic.

The system pipeline consists of four main stages: data collection and preprocessing, feature engineering, model training, and prediction with evaluation. Each stage is carefully designed to handle the challenges inherent in medical datasets, including missing values, class imbalance, and feature scaling. The preprocessing stage applies normalization and handles missing values using median imputation to ensure data quality.

3.1 Architecture Diagram



CHAPTER 4

MODULES DESCRIPTION

4.1 Data Collection and Preprocessing Module

This module is responsible for loading the Pima Indians Diabetes Dataset, which contains 768 patient records with 8 clinical attributes and a binary outcome variable. The dataset is publicly available from the UCI Machine Learning Repository. The preprocessing step involves identifying and handling missing values — several attributes such as glucose, blood pressure, insulin, and BMI contain zero values that are physiologically implausible and are treated as missing.

Missing values are imputed using the median value of the respective feature to preserve the dataset size. Feature scaling is applied using `StandardScaler` to normalize all features to a common scale, which is particularly important for distance-based algorithms such as SVM and K-Nearest Neighbors. The dataset is then split into training (80%) and testing (20%) sets using stratified sampling to maintain class distribution.

4.2 Model Training and Prediction Module

This module implements and trains multiple machine learning classification algorithms on the preprocessed dataset. The algorithms include Logistic Regression, which serves as the baseline model; Decision Tree Classifier, which uses entropy-based information gain for splitting; Random Forest Classifier, an ensemble method that aggregates multiple decision trees to reduce overfitting; and Support Vector Machine with a Radial Basis Function (RBF) kernel for non-linear classification.

Each model is trained using the training set and evaluated on the test set. Hyperparameter tuning is performed using Grid Search with 5-fold cross-validation to optimize model parameters. The best-performing model is selected based on accuracy and F1-score. The trained model is serialized using Python's pickle library for future deployment and integration into a web-based prediction interface.

CHAPTER 5

IMPLEMENTATION AND RESULTS

5.1 Experimental Setup

The implementation was carried out using Python 3.9 with the following libraries: NumPy and Pandas for data manipulation, Scikit-learn for machine learning algorithms and evaluation, Matplotlib and Seaborn for data visualization, and Jupyter Notebook as the development environment. The system was tested on a machine with an Intel Core i5 processor, 8 GB RAM, and Ubuntu 22.04 operating system.

The Pima Indians Diabetes Dataset was loaded from a CSV file. After preprocessing, 768 samples were used, with 500 non-diabetic cases (class 0) and 268 diabetic cases (class 1), reflecting a mild class imbalance. The SMOTE (Synthetic Minority Over-sampling Technique) oversampling method was applied during training to address this imbalance without losing data.

5.2 Results

The four models were evaluated on the held-out test set. The results are summarized below:

Table 5.1: Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	77.2%	75.4%	68.1%	71.5%
Decision Tree	74.5%	72.8%	65.3%	68.9%
Random Forest	82.4%	80.1%	76.8%	78.4%
Support Vector Machine	79.6%	77.2%	71.4%	74.2%

The Random Forest classifier outperformed all other models with an accuracy of 82.4% and an F1-score of 78.4%, indicating strong overall performance across both diabetic and non-diabetic classes. The feature importance analysis revealed that glucose concentration (28.3%), BMI (22.1%), and age (15.7%) were the three most significant predictors of diabetes. The confusion matrix showed that the Random Forest model correctly identified 76.8% of diabetic patients, which is critical for minimizing false negatives in a medical setting.

CHAPTER 6

CONCLUSION AND FUTURE WORK

This project successfully developed a machine learning-based diabetes prediction system capable of classifying patients as diabetic or non-diabetic based on clinical parameters. The system was trained and evaluated using the Pima Indians Diabetes Dataset, and four classification algorithms were implemented and compared. The Random Forest classifier achieved the best performance with an accuracy of 82.4%, demonstrating the effectiveness of ensemble learning methods for medical prediction tasks.

The project established that glucose level, body mass index, and age are the most influential features in predicting diabetes. The preprocessing pipeline, including median imputation and SMOTE oversampling, effectively addressed the challenges of missing values and class imbalance in the dataset. The results are consistent with findings reported in recent literature, validating the approach taken.

Future work could explore several avenues for improving the system. Deep learning approaches such as multilayer perceptrons and LSTM-based models could be explored for capturing temporal patterns in longitudinal patient data. Integration of additional clinical features such as HbA1c levels and dietary habits may improve predictive accuracy. The system could also be extended into a web-based or mobile application to facilitate real-time prediction for healthcare professionals. Furthermore, transfer learning from larger clinical datasets could be investigated to enhance generalization across diverse patient populations.

CHAPTER 7

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